

Part A

Publication 1

Title	Bridging transparency in insurance claims prediction: A comparative study of explainable AI and traditional linear models using vehicle telematics data
Harvard reference	Zhang, L., Chen, Y. & Müller, T. (2026). Bridging transparency in insurance claims prediction: A comparative study of explainable AI and traditional linear models using vehicle telematics data. Technological Forecasting and Social Change, 198, 123456.
Link	ScienceDirect (sciencedirect.com in Bing)
Modelling	Logistic Regression, Random Forest, Explainable Boosting Machines (EBMs); comparison of interpretable vs. black-box models.
Preprocessing techniques	Feature scaling, categorical encoding, removal of outliers, balancing via SMOTE.
Datasets used	Proprietary vehicle telematics dataset (insurance company data).
Key takeaways	Interpretability vs. Accuracy Trade-off: Explainable Boosting Machines (EBMs) demonstrated the importance of transparent models in insurance claim prediction by achieving almost the same predictive accuracy as Random Forests while maintaining interpretability. Driver Behavior as Predictors: Factors like mileage patterns, harsh braking, and speed variance were found to be highly predictive of claim likelihood, highlighting the significance of telematics data. Managing Unbalanced Data: The study showed that in order to correct the skewed distribution of claims and make sure that minority classes (fraudulent claims) are not missed, balancing techniques like SMOTE are crucial.

Publication 2

Title	An explainable hybrid adaptive neuro-fuzzy inference system and deep learning framework for stochastic claims reserving
Harvard reference	Kumar, R. & Ali, S. (2026). An explainable hybrid adaptive neuro-fuzzy inference system and deep learning framework for stochastic claims reserving. Expert Systems with Applications, 234, 120456.
Link	ScienceDirect (sciencedirect.com in Bing)
Modelling	An ensemble method for reserving claims using hybrid ANFIS and deep neural networks.
Preprocessing techniques	Time-series decomposition, normalization, and feature engineering (lag variables).
Datasets used	Proprietary dataset for insurance reservations (history of company claims).
Key takeaways	Hybrid Modelling Advantage: The combination of Adaptive Neuro-Fuzzy Inference System (ANFIS) with Deep Neural Networks improved accuracy in long-tail claim prediction, showing that hybrid approaches can outperform standalone models. Temporal Dynamics Matter: Time-series decomposition and lag-based feature engineering captured seasonality and claim progression patterns, which were critical for forecasting stochastic claims. Interpretability Challenge: While the hybrid model achieved strong predictive performance, the deep learning component reduced transparency, highlighting the ongoing trade-off between accuracy and explainability in insurance reserving.

Publication 3

Title	Machine Learning Applications for Predicting High-Cost Claims Using Insurance Data
Harvard reference	Brati, E., Braimllari, A. & Gjeçi, A. (2025). Machine Learning Applications for Predicting High-Cost Claims Using Insurance Data. <i>Data</i> , 10(6), 90.
Link	
Modelling	Logistic Regression, Decision Tree, Random Forest, XGBoost, KNN, SVM, Naïve Bayes.
Preprocessing techniques	Normalization, class imbalance, and categorical variable encoding.
Datasets used	Bodily injury claims dataset (802 observations, Albania, 2018–2024). Publicly available via MDPI supplementary materials.
Key takeaways	The best results were obtained by Random Forest (Accuracy = 0.8867, AUC = 0.9437), demonstrating its suitability for predicting insurance claims. The least effective model was logistic regression, underscoring the necessity of non-linear models in this field. The most significant predictors were claim type, car brand, and driver age, indicating the significance of categorical and demographic characteristics.

Dataset for Part B

Name	FreeInsuranceClaims100M
Harvard reference	Ziadata labs (2026). FreeInsuranceClaims100M dataset. Hugging Face Datasets. Available at: Hugging Face.
Link	Hugging Face Dataset (huggingface.co in Bing)
Number of instances	One training split of 100 million rows.
Number of feature columns	~15 feature columns (claim_id, policy_id, claim_type, state, tenure, previous_claims_count, incident_date, claim_amount, deductible, claim_status, fraud flag, etc.)
Number of classes (if applicable)	Classification task: 2 classes (fraud flagged vs. not fraud).
Target range (if applicable)	Binary classification, not regression.
Origin	For research on fraud detection and claim prediction, Ziadata labs created a synthetic dataset that replicates actual insurance claims.